Lab report

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Part 1 — Configure Network Devices

Router1

R1

Physical Config CLI Attributes

IOS Command Line Interface

```
R1(config)#interface g0/0
R1(config-if)#no shutdown

R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

R1(config-if)#interface g0/0.10
R1(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.10, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.10, changed state to up

R1(config-subif)#encapsulation dot1Q10
                                      ^
% Invalid input detected at '^' marker.

R1(config-subif)#encapsulation dot1Q 10
R1(config-subif)#ip address 192.168.10.1 255.255.255.0
R1(config-subif)#interface g0/0.20
R1(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.20, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.20, changed state to up

R1(config-subif)#encapsulation dot1Q 20
R1(config-subif)#ip address 192.168.20.1 255.255.255.0
R1(config-subif)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console
write memory
Building configuration...
[OK]
```

Switch1

S1

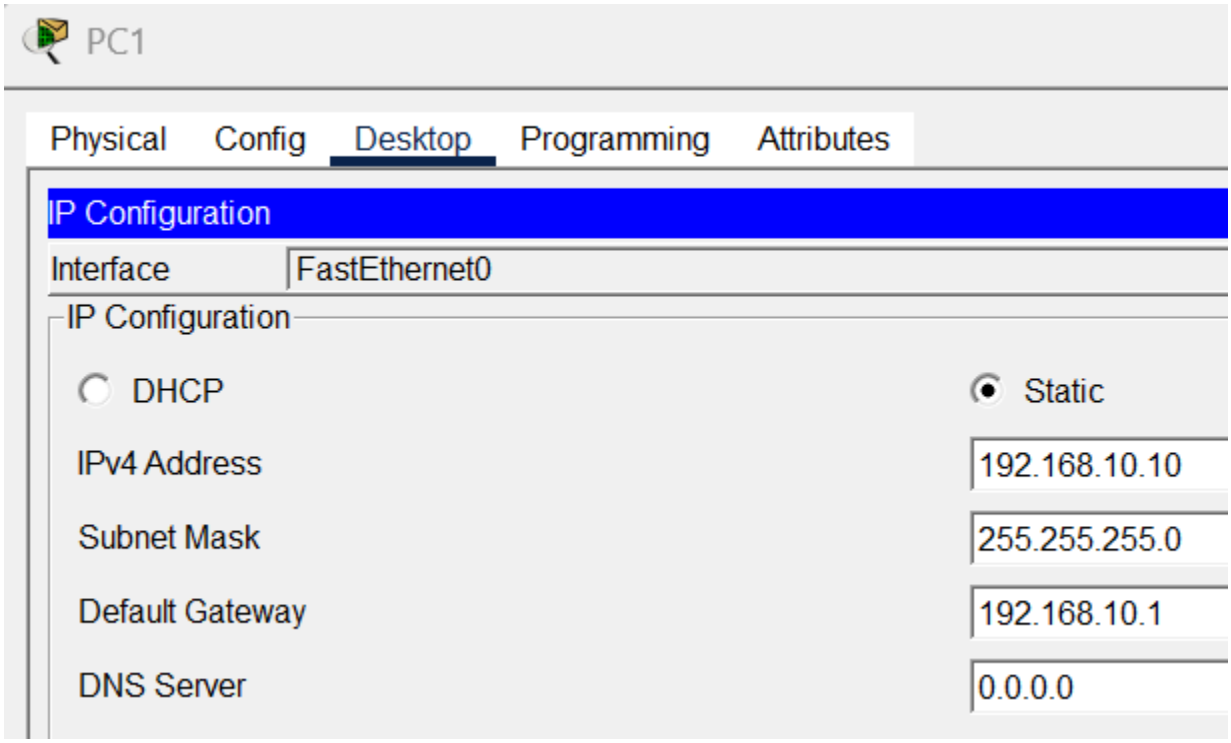
Physical Config CLI Attributes

IOS Command Line Inte

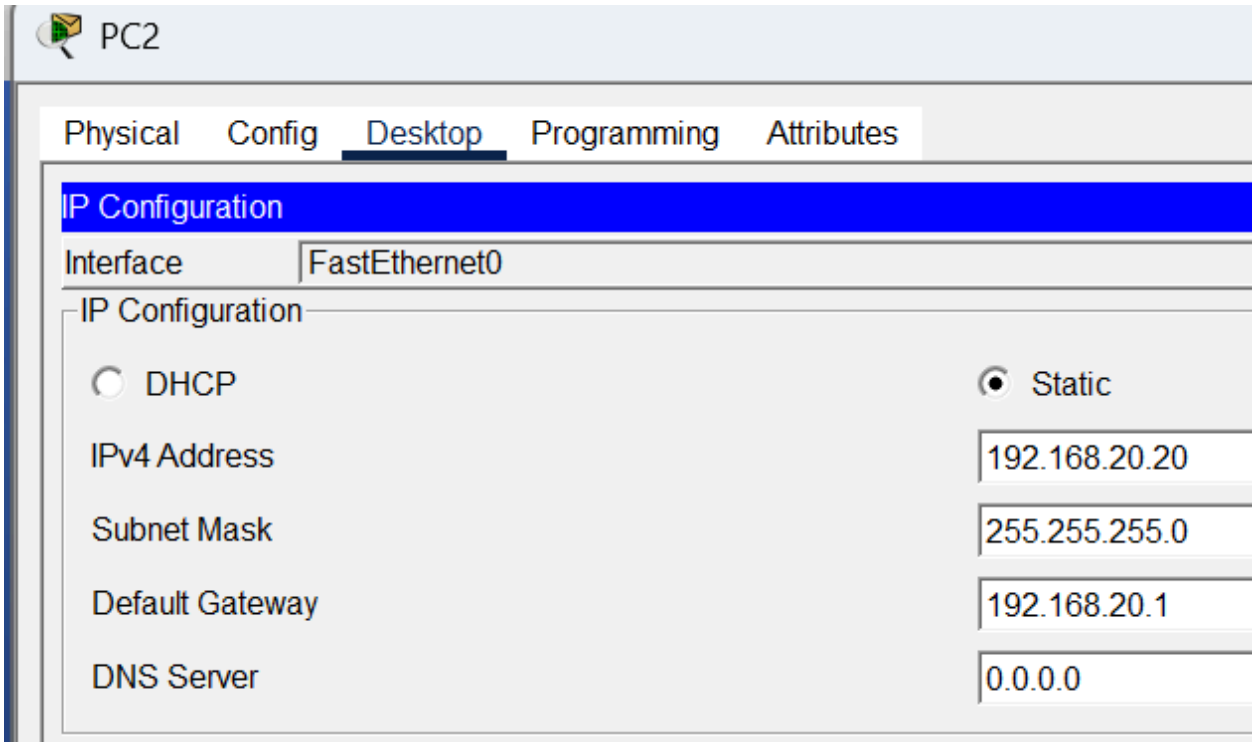
```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hostname S1
S1(config)#vlan 10
S1(config-vlan)#name CLIENTS
S1(config-vlan)#vlan 20
S1(config-vlan)#name SERVERS
S1(config-vlan)#
S1(config-vlan)#interface g0/1
S1(config-if)#switchport mode trunk
S1(config-if)#
S1(config-if)#interface fa0/1
S1(config-if)#switchport mode access
S1(config-if)#switchport access vlan 10
S1(config-if)#
S1(config-if)#interface fa0/2
S1(config-if)#switchport mode access
S1(config-if)#switchport access vlan 20
S1(config-if)#
S1(config-if)#end
S1#write memory
Building configuration...
[OK]
S1#
%SYS-5-CONFIG_I: Configured from console by console
```

Step 3 — Configure Hosts

PC1



PC2



Part 2 — Verify Connectivity

From PC1:
Ping 192.168.20.20 (PC2)

- Successful replies

```
C:\>ping 192.168.20.20

Pinging 192.168.20.20 with 32 bytes of data:

Reply from 192.168.20.20: bytes=32 time=30ms TTL=127
Reply from 192.168.20.20: bytes=32 time=10ms TTL=127
Reply from 192.168.20.20: bytes=32 time<1ms TTL=127
Reply from 192.168.20.20: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.20.20:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 30ms, Average = 10ms
```

- ARP entries

```
R1#show ip arp
Protocol  Address          Age (min)  Hardware Addr   Type   Interface
Internet  192.168.10.10           18  0001.97DA.6A15  ARPA   GigabitEthernet0/0.10
Internet  192.168.20.20           18  0001.4273.662D  ARPA   GigabitEthernet0/0.20
```

Router checks:

```
     192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.10.0/24 is directly connected, GigabitEthernet0/0.10
L       192.168.10.1/32 is directly connected, GigabitEthernet0/0.10
     192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.20.0/24 is directly connected, GigabitEthernet0/0.20
L       192.168.20.1/32 is directly connected, GigabitEthernet0/0.20
```

```
R1#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     unassigned      YES unset  up                    up
GigabitEthernet0/0.10  192.168.10.1    YES manual up                    up
GigabitEthernet0/0.20  192.168.20.1    YES manual up                    up
GigabitEthernet0/1     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
```

Switch checks:

```
S1#show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Gig0/1      on           802.1q         trunking      1

Port        Vlans allowed on trunk
Gig0/1      1-1005

Port        Vlans allowed and active in management domain
Gig0/1      1,10,20

Port        Vlans in spanning tree forwarding state and not pruned
Gig0/1      1,10,20
```

```
S1#show mac-address-table
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----

   1    0010.112b.9001    DYNAMIC     Gig0/1
  10    0010.112b.9001    DYNAMIC     Gig0/1
  20    0010.112b.9001    DYNAMIC     Gig0/1
```

🧪 Part 3 — Deploy MIME Server

On Server (PC2)

```
None
cd automation
python server/main_enhanced.py --host 0.0.0.0 --port 65432 --verbose
```

```
D:\KKU_files\year2_semester2\CP352005\LabOnSite\lab3\automation\week01-mime-typing>python server/main_enhance
d.py --host 0.0.0.0 --port 65432 --verbose
[18:48:05] INFO: Starting server with host=0.0.0.0, port=65432, storage=storage
[18:48:05] INFO: Server listening on 0.0.0.0:65432
```

On Client (PC1)

```
None
cd automation
python client/main_enhanced.py assets/sample.png --host 192.168.20.20 --port
65432
```

Confirm file saved.

```
D:\KKU_files\year2_semester2\CP352005\LabOnSite\lab3\automation\week01-mime-typing>python client/main_enhanced.py assets/sample.png --host 127.0.0.1 --
port 65432
[18:53:04] INFO: Starting client with 1 files to 127.0.0.1:65432
[18:53:04] INFO: Connected to 127.0.0.1:65432
[18:53:04] INFO: Sent assets/sample.png (108 bytes) as image/png
[18:53:04] INFO: Connection closed. Sent: 1, Failed: 0
[18:53:04] INFO: All files sent successfully

D:\KKU_files\year2_semester2\CP352005\LabOnSite\lab3\automation\week01-mime-typing>
```

📡 Part 4 — Wireshark Capture Walkthrough on PC1 and PC1

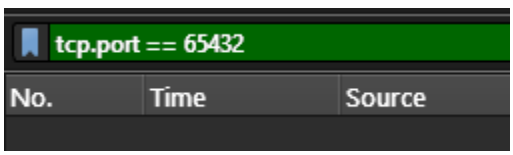
Step 1 — Start Capture

Open Wireshark.

Select interface connected to network.

Apply capture filter:

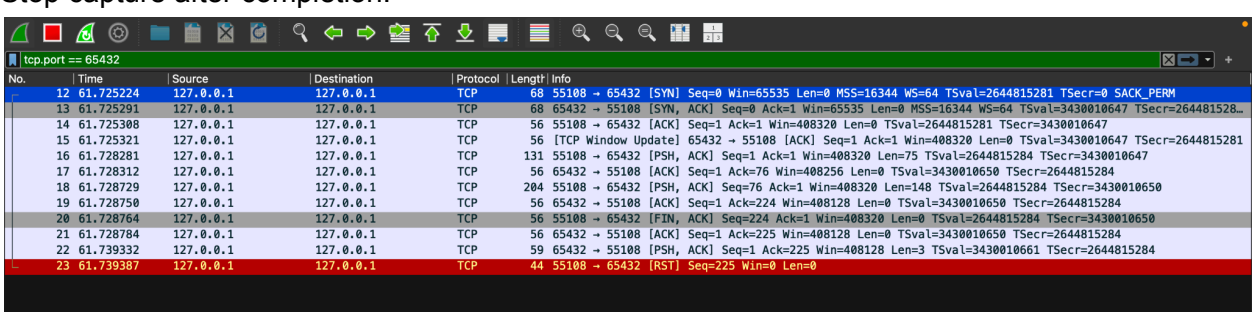
```
None
tcp port 65432
```



Step 2 — Initiate Transfer

Send file again.

Stop capture after completion.



Step 3 — Analyze Packets

A — ARP Exchange

Look for:

- Who has 192.168.20.20?

Confirm MAC resolution.

No ARP exchange was observed in this capture.

Reason:
The communication occurred over the loopback interface (127.0.0.1 to 127.0.0.1). Since both client and server were running on the same machine, no MAC address resolution was required. ARP is only used when communicating over a physical network segment.

B — TCP Three-Way Handshake

Identify:

1 SYN 2 SYN-ACK 3 ACK

Check sequence numbers.

1️⃣ SYN

Source port 58468 → Destination port 65432
Flag: SYN
Sequence number: 0

This packet initiates the connection.

2️⃣ SYN-ACK

Source port 65432 → Destination port 58468
Flags: SYN, ACK
Sequence number: 0
Acknowledgment number: 1

The server acknowledges the client's SYN and responds.

3️⃣ ACK

Source port 58468 → Destination port 65432
Flag: ACK
Sequence number: 1
Acknowledgment number: 1

The client confirms the connection.

Connection established successfully.

C — Data Transfer

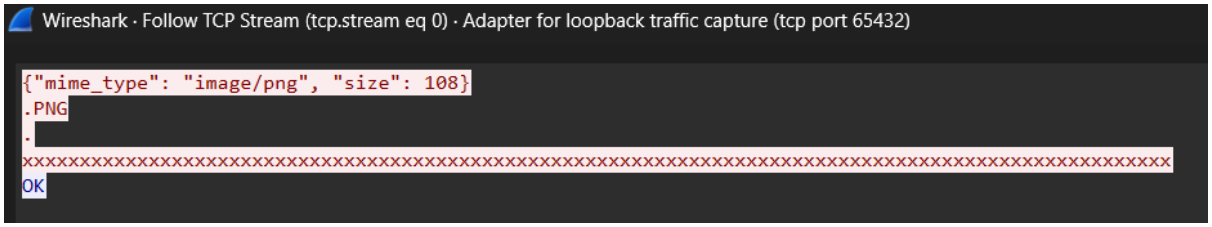
Follow stream:

Right click → Follow → TCP Stream.

You should see JSON header similar to:

```
None
{"mime_type":"image/png","size":12345}
```

Followed by binary.



D — VLAN Tags

If capturing on trunk:

Look for:

```
None
802.1Q Virtual LAN
```

Note VLAN ID.
No 802.1Q VLAN tags were observed in this capture.

Reason:

The packets were captured on the loopback interface (127.0.0.1). Loopback traffic does not use Ethernet framing and therefore does not include 802.1Q VLAN tagging.

🔬 Packet Dissection Guide

Layer	Field	What to Observe
Ethernet	Src/Dst MAC	Frame direction
802.1Q	VLAN ID	Segmentation
IP	TTL	Routing behavior
TCP	Flags	Reliability
Payload	JSON	MIME metadata

🛠️ Part 5 — Troubleshooting Labs

Scenario 1 — Broken Routing

Shutdown subinterface.

Observe:

- Ping failure
- TCP timeout

When the subinterface was shut down, inter-VLAN routing stopped. ICMP echo requests received no reply. TCP connection attempts resulted in timeouts because the router could not forward traffic between VLANs.

Routing decisions occur at Layer 3. When the interface was disabled, Layer 3 forwarding failed.

Scenario 2 — Firewall Block

Block port.

Observe:

- SYN retransmissions

When the application port was blocked, ICMP traffic still succeeded because routing was functional. However, TCP connection attempts showed repeated SYN retransmissions due to lack of response from the server.

TCP retransmits because it ensures reliable delivery. If no ACK is received, it assumes packet loss.

Scenario 3 — Wrong Gateway

Observe:

- ARP incomplete

When the default gateway was misconfigured, the client attempted ARP resolution for a non-existent IP address. No ARP reply was received, so no Layer 2 address could be determined. Without MAC resolution, communication could not proceed.

ARP occurs before IP routing because Layer 2 delivery must be established first.

Scenario 4 — Packet Loss Simulation

Drop packets.

Observe:

- TCP retransmissions
- Window adjustment

When packets were dropped, TCP detected missing sequence numbers and retransmitted the lost segments. Duplicate acknowledgments indicated gaps in received data. The TCP window size adjusted dynamically to maintain reliable and ordered delivery.

This demonstrates TCP's reliability and congestion control mechanisms.

Assignment Question

1. **Why does TCP retransmit lost segments?**
   => เพราะ TCP มีหน้าที่รับประกันความน่าเชื่อถือของการส่งข้อมูล (Reliability) เมื่อมีการส่งข้อมูล TCP จะรอรับข้อความยืนยัน ACK จากปลายทาง ถ้าไม่ได้รับ ACK ในเวลาที่กำหนด TCP (Timeout) จะถือว่าข้อมูลนั้นสสูญหาย แล้วจะทำการส่งใหม่อีกครั้ง

2. **How does VLAN tagging affect broadcast domains?**
   => VLAN tagging จะช่วยแบ่ง broadcast domain ออกจากกัน ซึ่งจะทำให้การสื่อสารระหว่างกันไม่ข้ามไป VLAN อื่นทำให้เครือข่ายไม่อืดและมีความปลอดภัยมากขึ้นและต้องใช้ Switch ในการข้ามไปยัง VLAN อื่น

3. **What role does MIME play compared to TCP headers?**
   => MIME ใช้ระบุชนิดและรูปแบบของข้อมูลที่ส่ง ใน Layer 7 (Application Layer) ส่วน TCP header ใช้ควบคุมการส่งข้อมูลให้ถึงปลายทางให้ถูกต้องและน่าเชื่อถือที่ Layer 4 (Transport Layer)

4. **Where does routing decision occur?**
   => การตัดสินใจ routing เกิดขึ้นที่ Network Layer (Layer 3) โดย router จะตรวจสอบ destination IP address ใน packet และเปรียบเทียบกับ routing table เพื่อเลือกเส้นทางและ interface ที่เหมาะสมในการส่ง packet ไปยังปลายทาง

5. **Why does ARP precede TCP communication?**
   => เพราะเนื่องจากเครื่องต้องทราบ MAC address ของปลายทางก่อน จึงจะสามารถส่ง packet ผ่านเครือข่ายได้ และการส่งข้อมูลจริงในเครือข่าย (Ethernet) ใช้ MAC address ใน Data Link Layer ดังนั้น ARP จึงทำหน้าที่แปลง IP address เป็น MAC address ก่อนเริ่มการสื่อสารด้วย TCP